

INTRODUCTION

Background

The first commercial versions, the SE555 (military-grade) and NE555 (commercial), were released by Signetics and developed by Hans Camenzind in 1972 and quickly became the only widely available monolithic timer IC on the market. The 555 timer operates in two primary modes, monostable and astable. Monostable mode sends out one output pulse for a fixed duration that goes high then stabilizes to low. The astable mode sends out continuous square waves, causing the circuit to oscillate from high to low .

Monostable Mode

In this mode, the circuit rests in a stable state with the output low and the discharge transistor on, keeping the capacitor discharged at 0V. When the trigger pin receives a low signal ($< 1/3 V_{cc}$), the flip-flop sets, pushing the output high and turning the discharge off. The capacitor then charges via the resistor until its voltage reaches $2/3 V_{cc}$. At this threshold, the flip-flop resets, the output drops low, the capacitor discharges, and the pulse is completed.

Astable Mode

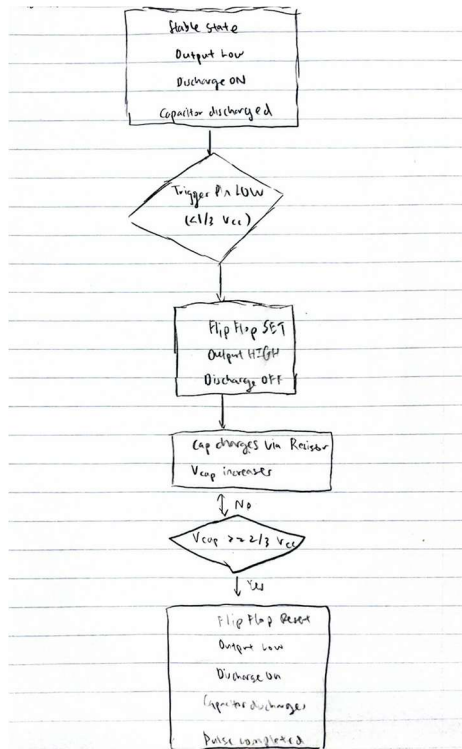
In astable mode, the 555 timer operates as an oscillator, meaning it lacks a stable state and continuously toggles the output between HIGH and LOW without requiring external triggering. This behavior is governed by the charging and discharging of the external capacitor through resistors R1 and R2 between two critical voltage thresholds: $1/3 V_{cc}$ and $2/3 V_{cc}$.

Objective and Interest

Explore monostable and astable modes of the LM555 chip, as well as analyzing the similarities and differences within the oscilloscope images, breadboard setup, and equations. Finally, the main goal of this project is to implement 555 timers into a real-life application (traffic light), and explore the practical challenges of tuning resistor and capacitor values to achieve the specific timing sequences required for safe traffic light operation.

THEORY

555 TIMER FLOWCHART:



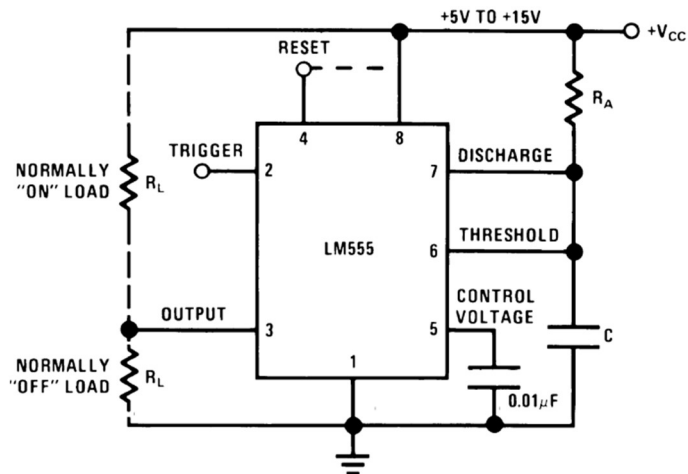
Monostable Mode of Operation

The Trigger: A negative-going pulse is applied to Pin 2. When this voltage drops below $1/3 V_{cc}$, the internal flip-flop sets.

The Output: The output (Pin 3) goes HIGH, and the internal discharge transistor (connected to Pin 7) turns off.

The Timing: With the discharge path closed, the external capacitor begins charging through resistor R_a . The voltage across the capacitor increases exponentially.

The Reset: When the capacitor voltage reaches $2/3 V_{cc}$, the internal threshold comparator (at Pin 6) resets the flip-flop. The output goes LOW, and the discharge transistor turns back on, instantly draining the capacitor to prepare for the next trigger.



Pin 1: Connected to Ground

Pin 2 (Trigger): Connected to the trigger source (button, wires touching, etc)

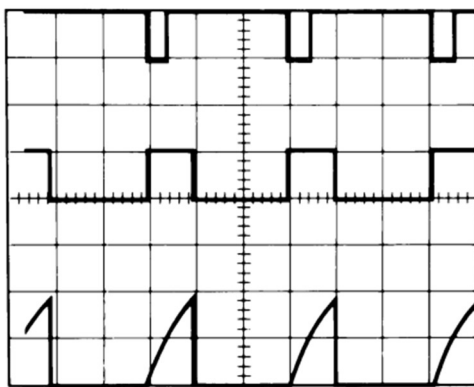
Pin 3 (Output): Connected to load (in our case, an LED)

Pin 4 (Reset): Tied to Vcc

Pin 5 (Control): Connected to a small capacitor to ground to filter noise.

Pin 6 & 7 (Threshold/Discharge): Tied together between the resistor R_A and capacitor C

Pin 8 (Vcc): Connected to the positive supply voltage



Top Waveform: Trigger Pin, sits at Vcc but is momentarily pulsed down to ground when the button is pushed.

Middle Waveform: Output Pin, connected to Pin 3, goes high when Trigger is low. Rises to a max of $\frac{2}{3} V_{cc}$ and then drops down to 0 V.

Lower Waveform: Pin 6, Capacitor, shows a sharkfin shaped curve, rises to $2/3 V_{cc}$ exponentially then drops to 0 V

KEY EQUATIONS:

$$\text{Pulse Width} = t = 1.1 * R * C$$

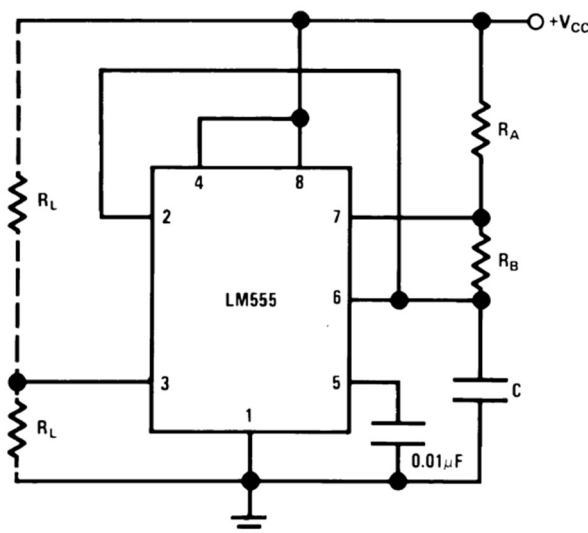
Astable Mode of Operation

The Charging Phase: The capacitor C charges up toward V_{cc} through both resistors in series, R_A and R_B . During this time, the output is HIGH.

The Threshold: When the voltage across the capacitor reaches $2/3 V_{cc}$, the internal comparator resets the flip-flop. The output (Pin 3) drops LOW, and the internal discharge transistor connected to Pin 7 turns on.

The Discharging Phase: The capacitor now discharges to ground through R_B and the discharge pin. Notice that it bypasses R_A entirely during the discharge phase.

The Trigger: When the capacitor voltage drops to $1/3 V_{cc}$, the trigger comparator sets the flip-flop again. The output goes HIGH, the discharge transistor turns off, and the cycle repeats automatically.



Pin 1: Connected to Ground

Pins 2 and 6 (Trigger and Threshold): Tied together. This allows the timer to continuously monitor the capacitor voltage for both the lower 1/3 Vcc lower limit and 2/3 Vcc upper limit.

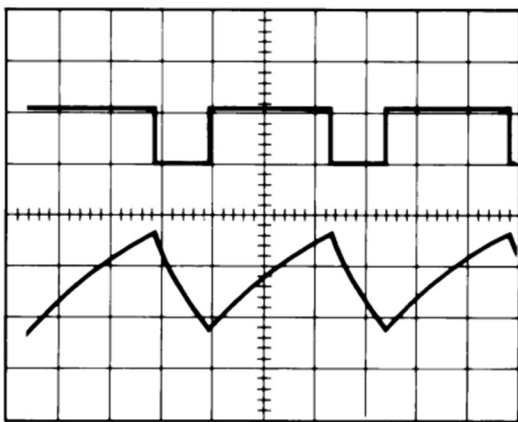
Pin 3: Output (LED)

Pin 4: Connected to Pin 8 (Vcc)

Pin 5: Tied to ground with 0.01 uF capacitor

Pin 7: Ra and Rb connected junction

Pin 8: Vcc



Top Waveform: Connected to output, High when charging, Low when discharging

Bottom Waveform: Connected to capacitor, oscillates between 1/3 Vcc and 2/3 Vcc.

KEY EQUATIONS:

$$\text{Duty Cycle} = D = (R_a + R_b) / (R_a + 2R_b)$$

$$\text{Time High} = t_h = 0.693 (R_a + R_b) * C$$

$$\text{Time Low} = t_l = 0.693(R_b) * C$$

$$\text{Total Time Period} = t = t_h + t_l = 0.693(R_a + 2R_b) * C$$

$$\text{Frequency} = f = 1.44 / [(R_a + 2R_b) * C]$$

MONOSTABLE MODE

EQUATIONS & CALCULATIONS

Monostable Mode:

$$t = 1.1 \cdot R \cdot C = 3 \text{ seconds}$$

Capacitor: use $100 \mu\text{F} = 1 \times 10^{-4} \text{ F}$

$$3 = 1.1 \cdot 10^{-4} \cdot R$$
$$R = \frac{3}{1.1 \cdot 10^{-4}}$$
$$= 27272.72 \Omega$$
$$\approx \underline{27 \text{ k}\Omega} \quad [\text{nearest resistor value}]$$

Expected Time: 3 seconds

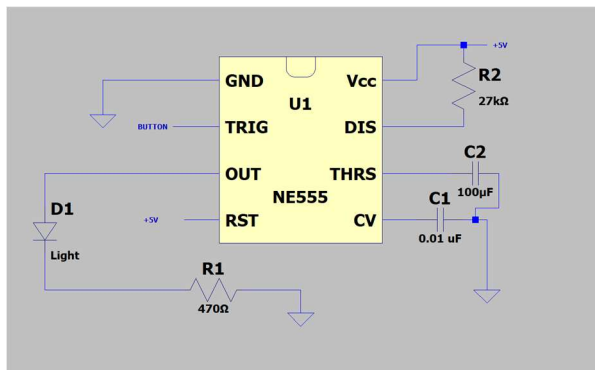
Calculated after using nearest resistor value:

$$t = 1.1 \cdot 27 \text{ k}\Omega \cdot 10^{-4} \text{ F} = \underline{2.97 \text{ seconds}}$$

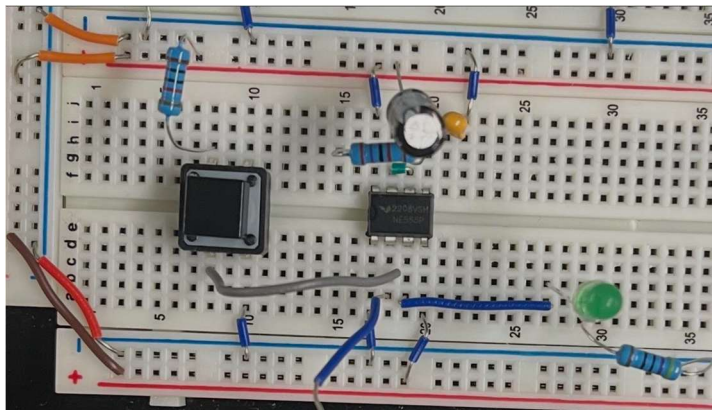
Actual time on breadboard: 3.21 seconds

$$\left(\frac{3.21 - 2.97}{2.97} \right) \times 100\% = \underline{8.08\% \text{ error}}$$

SCHEMATIC



BREADBOARD



OSCILLOSCOPE IMAGE



Actual Time: 3.21 seconds

ASTABLE MODE

EQUATIONS & CALCULATIONS

60% Duty Cycle (2 second period)

Astable Mode 60% Duty Cycle 2 second period

Duty Cycle:

$$D = \frac{R_A + R_B}{R_A + 2R_B} = 0.6$$

$$0.6 R_A + 1.2 R_B = R_A + R_B$$

$$0.2 R_B = 0.4 R_A$$

$$R_B = 2 R_A$$

Frequency of oscillation:

$$\frac{1}{T} = \frac{1}{(R_A + 2R_B)C}$$

$$T = 0.693 \cdot (R_A + 2R_B) \cdot C$$

$$T = 0.693 (R_A + 4R_A) \cdot C = 0.693 (5R_A) \cdot C$$

$$\frac{2}{0.693 \cdot 5} = R_A \cdot C = 0.5772$$

Use 100 μF capacitor: $R_A = \frac{0.5772}{10^{-4}} = 5772 \Omega$ $R_A = 5772 \Omega$ $C = 100 \mu F$

$$R_B = 2R_A = 5772 \Omega \cdot 2 = 11544 \Omega$$
 $R_B = 11544 \Omega$

Actual values used:

$$R_A = 50 k\Omega$$
 $R_B = 100 k\Omega$ $C = 10 \mu F$

$$T = 0.693 \cdot (50000 + 200000) \cdot 10^{-5} = 2.051 s$$

$$\left(\frac{12.510 - 2.051}{2.051} \right) \times 100\% = 0.54\%$$

C2 capacitor is supposed to be 10 uF, equation uses 10⁻⁵, but wrote on right-hand side 10⁻⁴.

75% Duty Cycle (1 second period)

As stable Mode : 75% Duty cycle

$$D = \frac{R_A + R_B}{R_A + 2R_B} = 0.75$$

$$0.75 R_A + 1.5 R_B = R_A + R_B$$

$$0.5 R_B = 0.25 R_A$$

$$2 R_B = R_A$$

Frequency of oscillation

$$\frac{1}{T} = \frac{1.44}{(R_A + 2R_B)C}$$

$$T = 0.693 (R_A + 2R_B) \cdot C$$

$$1 = 0.693 (R_A + 2R_B) \cdot 10^{-5}$$

$$1 = 0.693 (4R_B) \cdot 10^{-5}$$

$$R_B = \frac{1}{0.693 \cdot 4 \cdot 10^{-5}}$$

$$R_B = 36075.03 \quad \text{Chosen } R_B = 39000$$

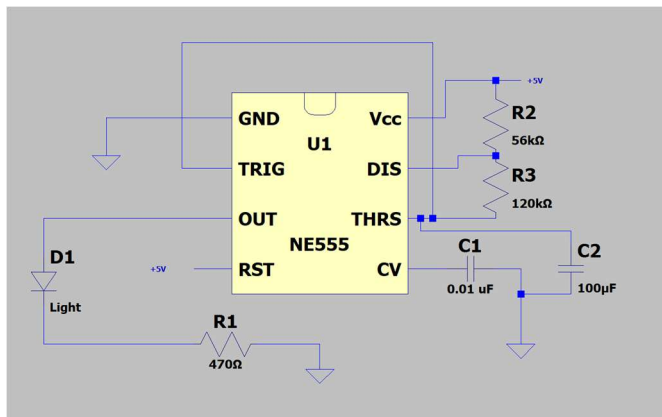
$$R_A = 72150.07 \quad \text{Chosen } R_A = 68000$$

$$0.693 (68000 + 2(39000)) \cdot 10^{-5} = 1.012 \text{ s}$$

$$\left(\frac{1.012 - 1.012}{1.012} \right) \times 100\% = 11.26\%$$

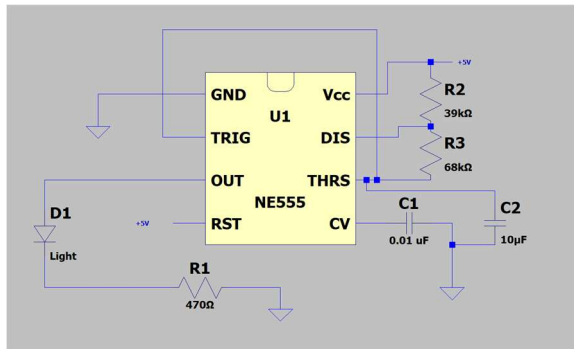
SCHEMATIC

60% Duty Cycle



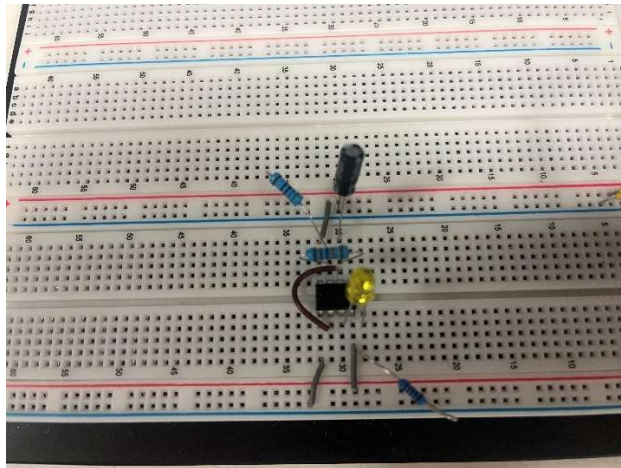
C2 capacitor is supposed to be 10 μF

75% Duty Cycle

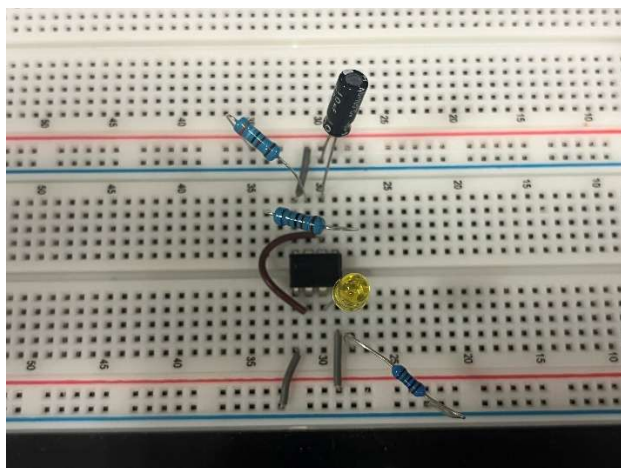


BREADBOARD

60% Duty Cycle

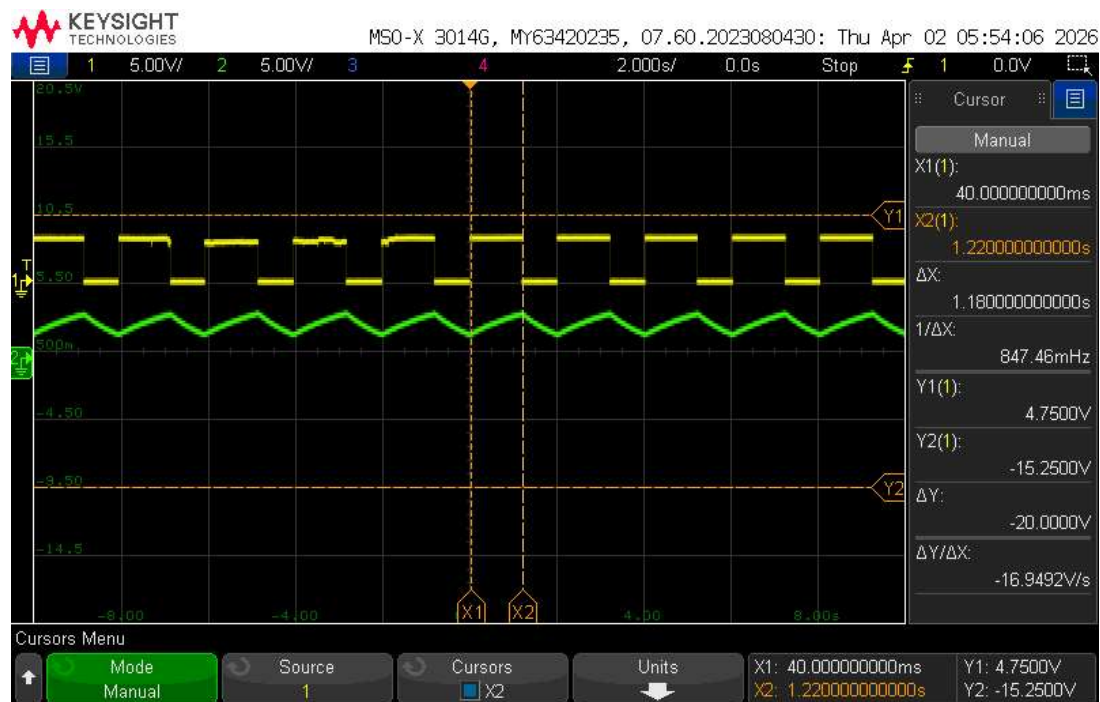


75% Duty Cycle

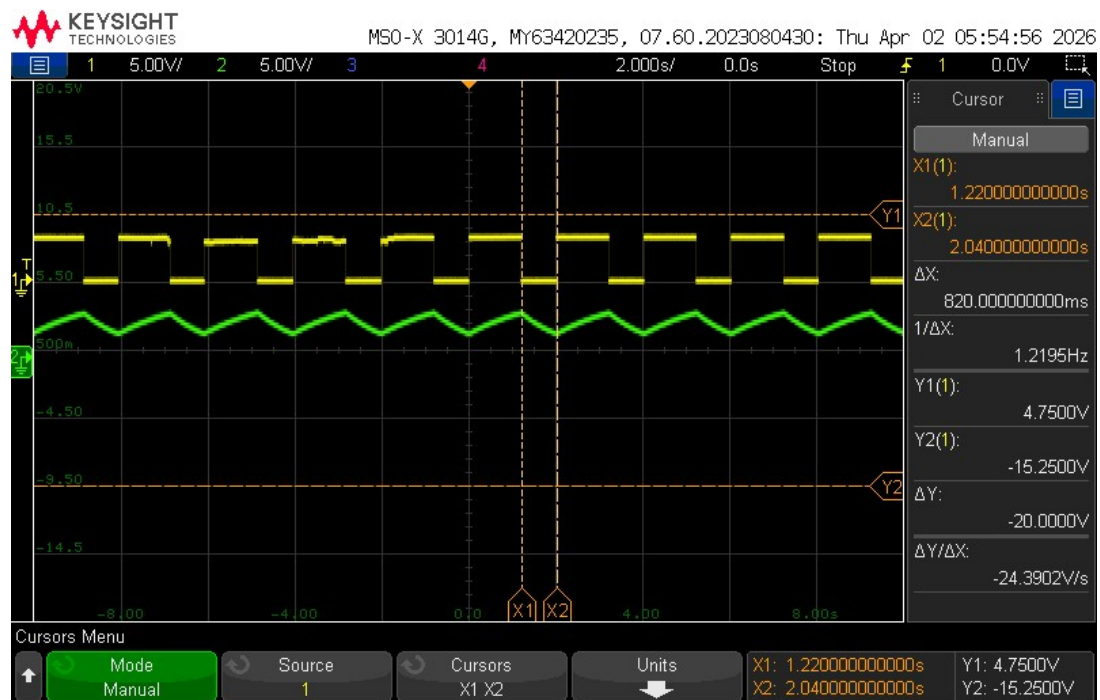


OSCILLOSCOPE IMAGE

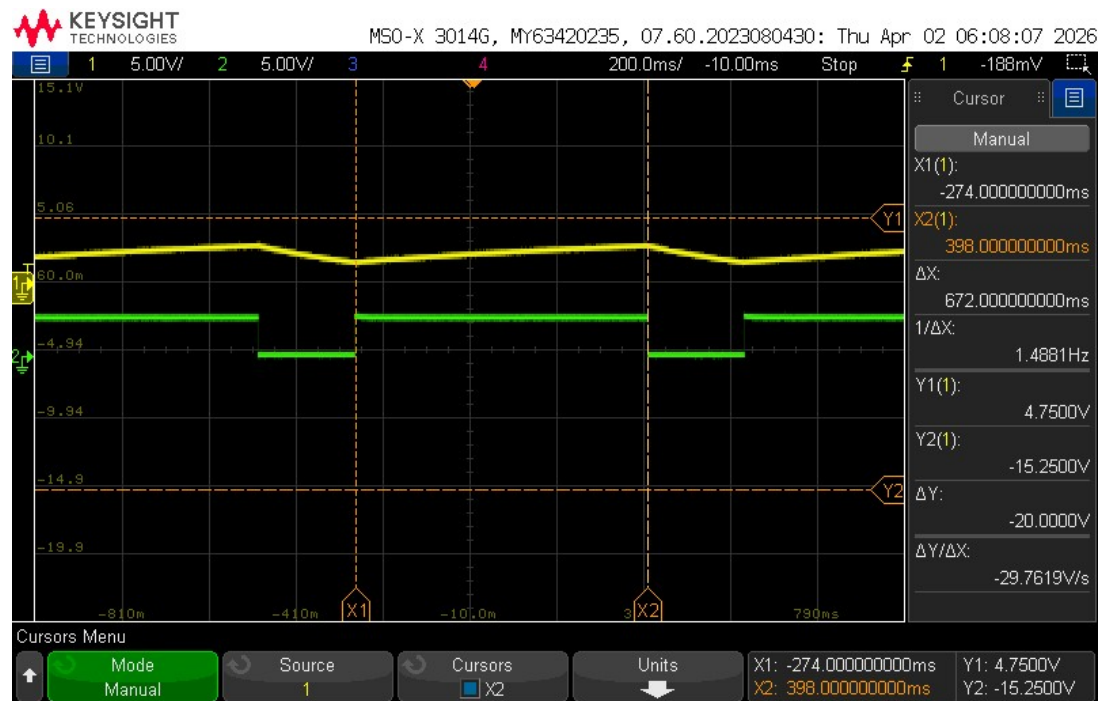
60% duty cycle (rise)



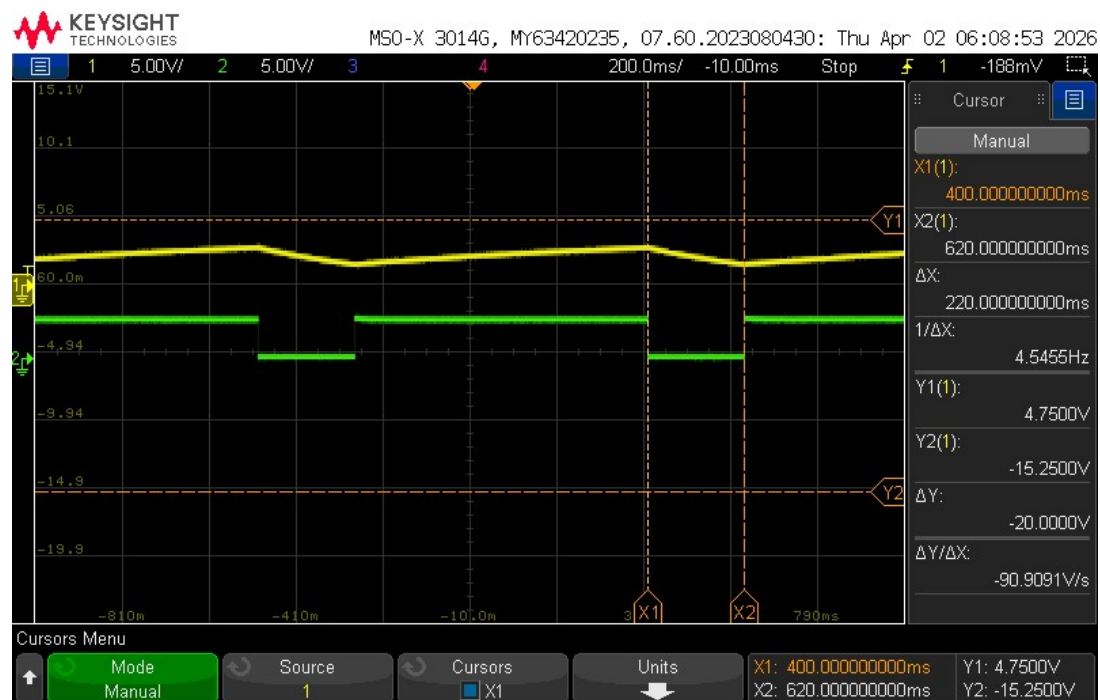
60% duty cycle (fall)



75% duty cycle (rise)



75% duty cycle (fall)



MAIN PROJECT: TRAFFIC LIGHT

Why build a traffic light?

Before there were sensors to determine whether a car or pedestrian was on the street, humans manually controlled the lights, using the green-yellow-red configuration we use today to communicate to those on the road. Eventually, engineers implemented mechanical timers (For example, some streets would use a 30 second green light, 5 second yellow light, and 30 second green light) so that manual labor was not required. I wanted to create a circuit that simulates these mechanical timers using two LM555 IC's along with other electrical components. Also, because timers are involved, I thought that it would be good to develop a pattern of oscillating signals to automatically sequence the different phases in a continuous loop.

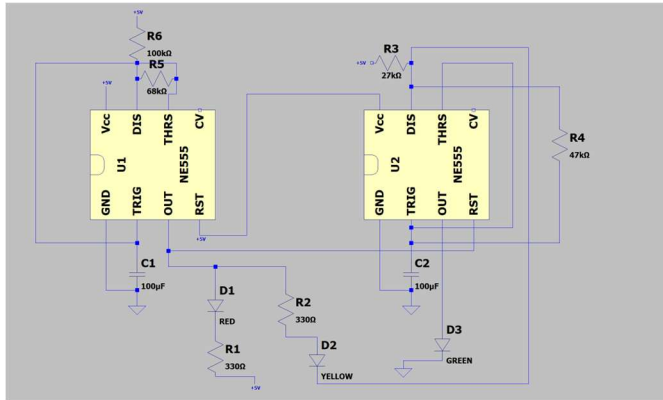
CIRCUIT OPERATION

Using the resistors, we find the amount of time each light is on for (astable mode equations). The green light is turned on, then is turned off when the yellow light is turned on, then the red light is turned on when the yellow light is turned off, and this is one period. The entire period lasts around 12 seconds, with each light being turned on for the same amount of time every single cycle.

EQUATIONS & CALCULATIONS

Traffic Light Calculations
Green Light:
Left Timer resets every cycle:
$t = 1.1 \cdot (21 \text{ k}\Omega + 47 \text{ k}\Omega) \cdot 1.0 \cdot 10^{-4} \text{ F}$
$= 8.1 \text{ seconds}$
Green Light should stay on for around 8.1 seconds
Yellow Light:
$t = 0.693 \cdot 47 \text{ k}\Omega \cdot 1.0 \cdot 10^{-4} \text{ F}$
$= 3.2 \text{ seconds}$
Yellow Light should stay on for around 3.2 seconds
Red Light:
$t = 0.693 \cdot 68 \text{ k}\Omega \cdot 1.0 \cdot 10^{-4} \text{ F}$
$= 4.7 \text{ seconds}$
Red Light should stay on for around 4.7 seconds
Average Window:
$t = 0.693 \cdot (100 \text{ k}\Omega + 68 \text{ k}\Omega) \cdot 1.0 \cdot 10^{-4} \text{ F}$
$= 11.6 \text{ seconds}$

SCHEMATIC



Connections:

Left Pin 1: Ground

Left Pin 2: 100 uF capacitor → ground, pin 6

Left Pin 3: right pin 4, red light → 330 ohm → ground, 330 ohm → yellow light → right pin 7

Left Pin 4: right pin 8, Vcc

Left Pin 5: None

Left Pin 6: 68 kohm resistor → pin 7, pin 2

Left Pin 7: 68 kohm resistor → pin 6, 100 kohm resistor → ground

Left Pin 8: Vcc

Right Pin 1: Ground

Right Pin 2: 100 uF capacitor → ground, pin 6, 47 kohm resistor → pin 7

Right Pin 3: Green light → ground

Right Pin 4: pin 3

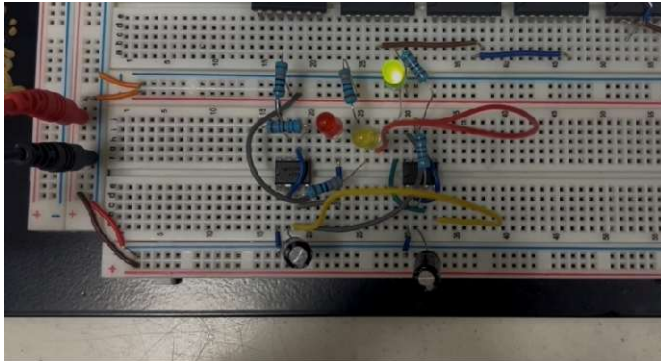
Right Pin 5: None

Right Pin 6: pin 2

Right Pin 7: 47 kohm → pin 2, 27 kohm → Vcc

Right Pin 8: left pin 4

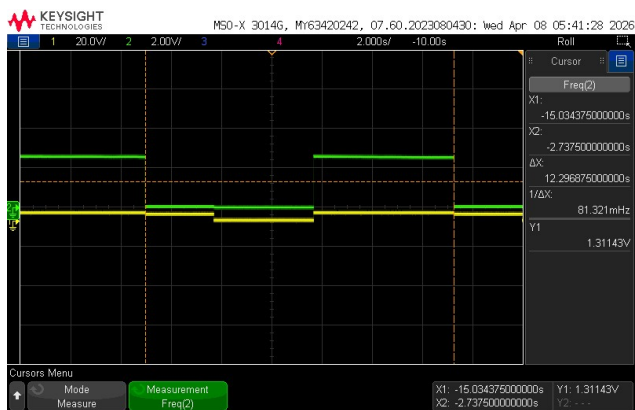
BREADBOARD



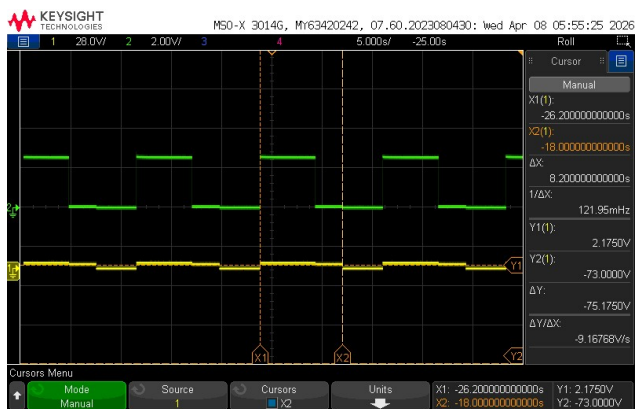
Attached is a photo, but I have a video that shows the entire period (green → yellow → red)

OSCILLOSCOPE

Total Period

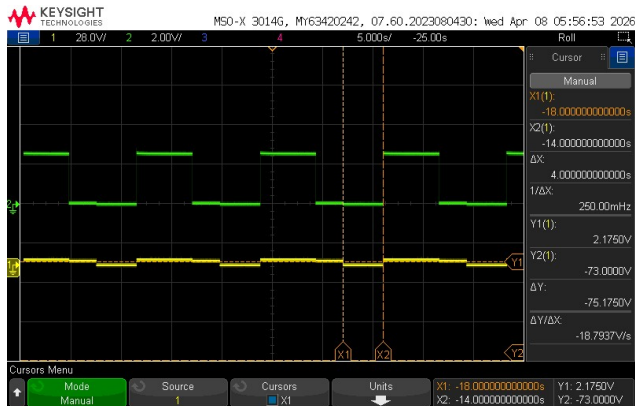


Green Phase



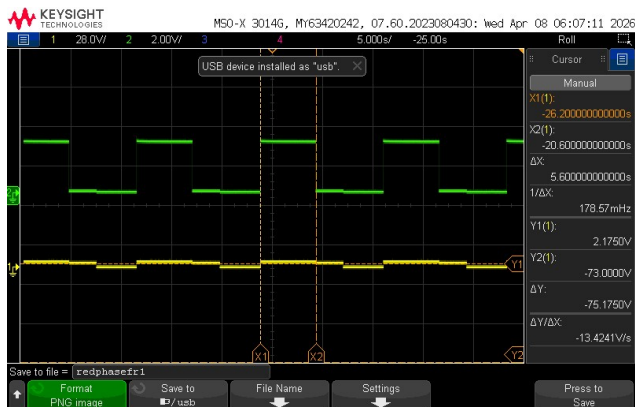
DeltaX = 8.2 s

Yellow Phase



DeltaX = 4.00 s

Red Phase



DeltaX = 5.6 s

RESULTS & DISCUSSION

MONOSTABLE

	Time (s)	R_a (ohms)	C (uF)	Percent Error for Time
Expected	3.00	27272.27	100	-
Actual (changed resistance)	2.97	27k	100	-
Actual (shown on oscilloscope)	3.21	27k	100	8.08%

POSSIBLE DEVIATIONS

- Internal resistance of the wires
- Oscilloscope issues (technology)
- Capacitance tolerance and leakage

ASTABLE (60% DUTY CYCLE)

	Time (s)	R_a(ohms)	R_b(ohms)	C (uF)	Percent Error for Time
Expected	2.00	115440	57720	10	-
Actual (changed resistance)	2.051	120k	56k	10	-
Actual (shown on oscilloscope)	2.010	120k	56k	10	0.54%

POSSIBLE DEVIATIONS

- Internal resistance of the wires
- Tolerance of the resistors
- Oscilloscope measurement issues

ASTABLE (75% DUTY CYCLE)

	Time (s)	R_a(ohms)	R_b(ohms)	C (uF)	Percent Error for Time
Expected	1.00	72150	36075	10	-
Actual (changed resistance)	1.012	68k	39k	10	-
Actual (shown on oscilloscope)	0.892	68k	39k	10	11.86%

POSSIBLE DEVIATIONS

- Internal resistance of the wires
- Tolerance of the resistors
- Oscilloscope measurement issues

CONCLUSION

This project successfully demonstrated the application of 555 timers in a two-chip configuration to model a real-world traffic light, providing valuable hands-on experience in both system design and practical troubleshooting. By utilizing astable operating modes and calculating specific RC time constants, my circuit was successfully tuned to achieve custom, asymmetrical timing for the green, yellow, and red phases. However, mapping the high and low states of the square waves with an oscilloscope revealed the significant gap between ideal schematics and physical circuit behavior. To improve the results of the experiment, I can build the circuit more than once with different resistors because I could have faulty equipment, and performing an experiment multiple times will allow me to get a better result. Overcoming real-world hardware limitations such as rapid strobing from capacitor mismatches, race conditions caused by component tolerances, and implementing a 'seesaw' logic state to prevent overlapping lights reinforced a critical engineering concept: precise digital logic relies entirely on the careful tuning of the underlying analog components.

Sources:

[1] *LM555 Timer*, Texas Instruments, Dallas, TX, USA, 2015. [Online]. Available: <https://www.ti.com/lit/ds/symlink/lm555.pdf>. [Accessed: Apr. 10, 2026].

[2] W. Storr, "555 Timer Monostable Circuit," *Basic Electronics Tutorials*. [Online]. Available: <https://www.electronics-tutorials.ws/waveforms/555-monostable.html>. [Accessed: Apr. 10, 2026].

[3] W. Storr, "555 Timer Astable Oscillator," *Basic Electronics Tutorials*. [Online]. Available: https://www.electronics-tutorials.ws/waveforms/555_oscillator.html. [Accessed: Apr. 10, 2026].